

1. Minimize the Boolean function: $F(A,B,C,D) = A'B'C' + A'BC'D + AB'C' + ABC'D + C$

(A) $C + C'$

(B) $C' + D$

(C) $C + B' + D$

(D) $C + C'D$

2. How many essential prime implicants are present in $F(A,B,C,D) = \sum m(0, 2, 3, 5, 7, 8, 10, 11, 14, 15)$?

(A) 3

(B) 2

(C) 4

(D) 5

3. Simplify the expression using consensus theorem: $F = AB + A'C + BC + BD + A'D$

(A) $AB + A'C + A'D$

(B) $AB + A'C + BC$

(C) $AB + BC + A'D$

(D) $A'C + BD + AB$

4. Find the minimal NOR-NOR circuit realization for $F(A,B,C) = \sum m(1, 2, 3, 7)$.

(A) $(A' + B') + (A + C)'$

(B) $[(A' + C)' + (A + B')' + (B + C)']'$

(C) $[(A + B)' + (A' + C')']'$

(D) $(A + B + C)'$

5. Identify the self-dual function among the following 3-variable logic expressions:

(A) $AB + BC + CA$

(B) $A + B + C$

(C) $ABC + A'B'C'$

(D) $A \oplus B \oplus C$

6. What is the total number of neutral functions (equal 0s and 1s) possible for n variables?

(A) 2^{2^n}

(B) 2^{n-1}

(C) $C(2^n, 2^{n-1})$

(D) $2^{2^{n-1}}$

7. Eliminate static 1-hazards in $F = A \cdot C + B \cdot C'$ by adding which redundant prime implicant?

(A) $A \cdot B$

(B) $A' \cdot B'$

(C) $A \cdot B'$

(D) $A' \cdot B$

8. Simplify $F(A,B,C,D) = \prod M(0, 1, 4, 5, 8, 9, 12, 13)$ into minimal POS form:

(A) $C + D$

(B) $C' + D'$

(C) $A + B$

(D) $(A + C) \cdot (B + D)$

9. Minimum 2-input NOR gates required to implement a 2-to-1 Multiplexer is:

(A) 4

(B) 3

(C) 5

(D) 6

10. Simplify: $F = (A \oplus B) \odot (B \oplus C) \odot (C \oplus A)$ for 3 binary variables.

- (A) 1 (B) 0
(C) $A \oplus B \oplus C$ (D) $ABC + A'B'C'$

11. Given $F(A,B,C) = \sum m(1, 3, 6, 7)$ with don't cares $d(A,B,C) = \sum m(0, 5)$, minimal SOP is:

- (A) $C + AB$ (B) $C' + AB$
(C) $B + AC$ (D) $A + BC$

12. Total number of self-dual Boolean functions of n variables is given by:

- (A) $2^{2^{n-1}}$ (B) $2^{2^n - 1}$
(C) 2^{n-1} (D) 2^{2^n}

13. Convert the expression $F = A'B + AC' + BC$ into minimal product of sums (POS):

- (A) $(A+B) \cdot (A'+C') \cdot (B+C')$ (B) $(A+B') \cdot (A'+C)$
(C) $(A'+B) \cdot (A+C')$ (D) $(A+B+C) \cdot (A'+B'+C')$

14. Reduce: $F = (X + Y + Z) \cdot (X + Y + Z') \cdot (X + Y' + Z) \cdot (X' + Y + Z)$

- (A) $(X+Y) \cdot (X+Z) \cdot (Y+Z)$ (B) $X+Y+Z$
(C) $XY+YZ+ZX$ (D) $X'+Y'+Z'$

15. A circuit has output $F = (A \oplus B) \oplus (C \oplus D)$. Output is 1 if and only if:

- (A) An odd number of inputs (A,B,C,D) are 1 (B) An even number of inputs are 1
(C) Exactly two inputs are 1 (D) All inputs are identical

16. Find dual of complement of $F = A \cdot B' + A' \cdot B$:

- (A) $A \cdot B' + A' \cdot B$ (B) $(A+B') \cdot (A'+B)$
(C) $AB + A'B'$ (D) $A'B' + AB$

17. Determine the number of prime implicants for $F(W,X,Y,Z) = \sum m(0, 1, 2, 5, 8, 9, 10)$:

- (A) 3 (B) 4
(C) 5 (D) 2

18. Boolean operator $A \odot B$ can be represented solely with 2-input NAND gates using a minimum of:

- (A) 5 gates (B) 4 gates
(C) 6 gates (D) 3 gates

19. Simplify the expression: $F = (A + B'C)' + (A'B + C')' + A'B'C'$

- (A) $A'(B + C')$ (B) $A'B'$
(C) $A'C'$ (D) $A'(B' + C')$

20. If $F(A, B, C) = A \oplus B \oplus C$, its maxterm representation is:

- (A) $\prod M(0, 3, 5, 6)$ (B) $\prod M(1, 2, 4, 7)$
(C) $\prod M(0, 1, 2, 3)$ (D) $\prod M(4, 5, 6, 7)$

21. How many literal appearances remain in fully minimized SOP form of $F = A'BC + AB'C + ABC' + ABC$?

- (A) 6 (B) 4
(C) 3 (D) 8

22. Minimal NAND implementation of $F(A, B, C) = \sum m(0, 6)$ with don't cares $d = \sum m(1, 2, 4)$ requires:

- (A) 1 inverter + 1 gate (B) 3 NAND gates
(C) 2 NAND gates (D) 4 NAND gates

23. Simplify: $F = (A + B)(A + C)(B + C')$

- (A) $AC + BC'$ (B) $AB + AC'$
(C) $A + BC'$ (D) $B(A + C')$

24. Which function is functionally complete by itself?

- (A) $F(A, B) = A'B'$ (B) $F(A, B) = A \oplus B$
(C) $F(A, B) = AB$ (D) $F(A, B) = A + B$

25. In a 4-variable K-map, a octet (group of 8 adjacents) reduces to a product term containing:

- (A) 1 literal (B) 2 literals
(C) 3 literals (D) 4 literals

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(C)	6	(C)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(C)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(B)	9	(C)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)